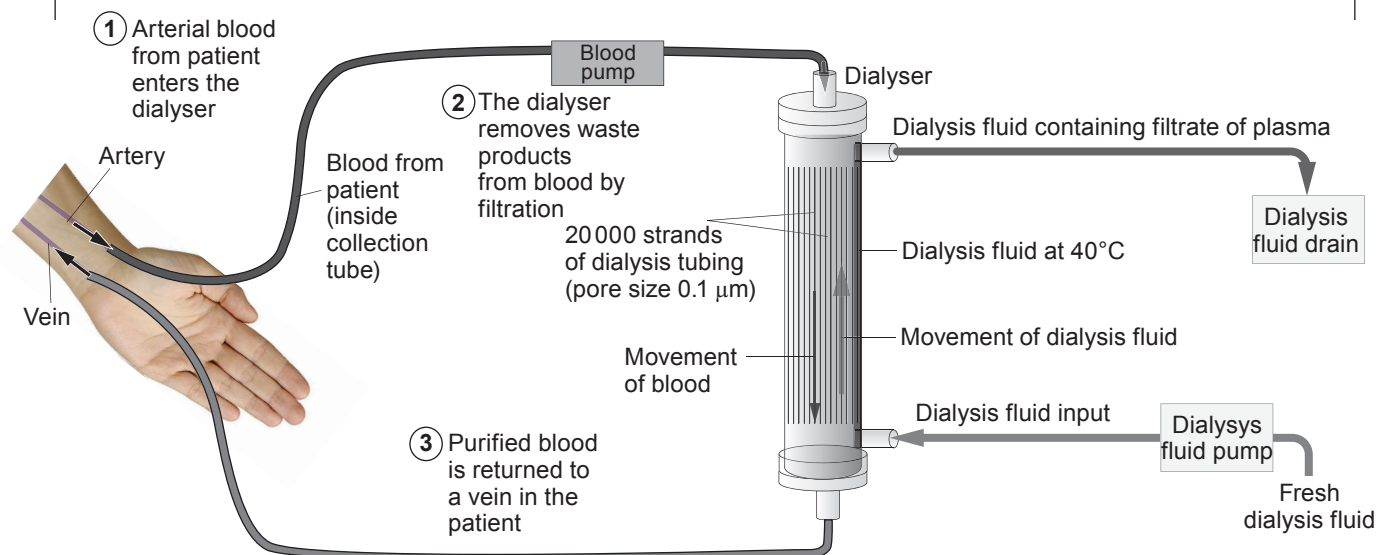


3. When the kidneys are damaged by disease they may not be able to excrete urea or control the water potential of the blood plasma.

Patients may have to use an artificial kidney (dialysis machine). This will carry out the filtration of molecules from the blood.

Blood is pumped from the patient and is passed through tubes formed from selectively permeable cellulose acetate membranes. The other side of the membrane has sterile dialysis fluid continually pumped over it.



- (a) With reference to the diagram above, describe and explain how the dialysis machine makes filtration more efficient. [4]

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Aquaporin is a type of protein channel that allows water movement through the membranes of the epithelial cells in the collecting duct. Antidiuretic hormone (ADH) is a protein made in mammals. It consists of a chain of nine amino acids.

- (i) State the minimum number of nucleotides on the DNA template strand that code for ADH. [1]

.....



Diabetes insipidus is a condition that results in excessive thirst and production of large volumes of dilute urine. This condition is different from diabetes mellitus which may result in increased glucose concentration in the urine.

One cause of diabetes insipidus is the patient not releasing ADH.

(c) Suggest how this causes diabetes insipidus. [1]

.....

.....

Another form of diabetes insipidus is nephrogenic diabetes insipidus. This is a genetic condition caused by a dominant allele (N). This results in a change in the shape of the receptor sites for ADH.

(d) A heterozygous sufferer with this condition has a partner who is a non-sufferer. Draw a suitable genetic diagram and use this to determine the probability of them having a child who would suffer from nephrogenic diabetes insipidus. [3]

probability =

15

